Same patient, different number

Work in pairs. Where you see ✎ **Predict**, write your answer **before** you click. Do the blocks in the order **A, B, C**. The task numbers match the tasks on the web pages; **Bonus** tasks are for fast pairs.

patrikbrynolfsson.github.io/estro2026-dwi-handson

Names _____ Specialty _____

A Has the tumour responded? · do first

Prostate, two scans 13 days apart

2 Expert ROI drawn on ADC: mean ADC on each scan.

Scan 1 _____ Scan 2 _____ × 10^{-3} mm^2/s Change _____ %

3 Change the ROI. How much does the value move **within scan 1**?

ROI on scan 1	Mean	Median	10th percentile
Expert, on ADC			
Grown 2 mm			
Shrunk 2 mm			
Expert, on T2w			

Moves least: ☐ Mean ☐ Median ☐ 10th percentile

4 ✎ **Predict** Scan 1 was before treatment, scan 2 two weeks in. Has the tumour responded? Why?

☐ Response ☐ No response ☐ Can't tell

5 **Bonus** Does your conclusion depend on which ROI you used?

B Contouring on DWI · second

Another man from the same study · numbers = the four steps

1 Your outline on T2w:

Volume _____ cm^3

2 ✎ **Predict** On DWI it will be

☐ bigger ☐ same ☐ smaller

Your DWI volume _____ cm^3

3 Compare your two outlines, then flick T2w ↔ DWI and measure the shift at the back of the prostate with the ruler.

Prediction right? ☐ Yes ☐ No Overlap (Dice) _____ Centre shift _____ mm

Ruler _____ mm

4 The radiologist: overlap (Dice) _____ DWI outline further back by _____ mm

DISCUSS

1 Would you use the DWI outline as the GTV, as a boost volume (as in FLAME), or only as a guide when you contour on T2w?

☐ GTV ☐ Boost volume ☐ Guide only

2 Margin that would cover the shift you measured: _____ mm

3 Would distortion correction or registration remove the shift? What would you check?

Prostate data: QIN-PROSTATE-Repeatability, TCIA, doi:10.7937/K9/TCIA.2018.MR1CKGND (CC BY 4.0); Fedorov et al., Sci Data 2018;5:180281 and Invest Radiol 2017;52:538-46 (tumour ADC limit 42 %; volume 71 % on T2w, 112 % on ADC). FLAME: Kerkmeijer et al., J Clin Oncol 2021;39:787-96.

C Same brain, different ADC · third, after the break

TAB 1 · SAME BRAIN, DIFFERENT ADC

1 Move the b-value slider from 0 to 1000. Which parts of the brain lose the most signal between b = 0 and b = 100?

_____

2 Brain tissue: read its ADC with each choice of b-values.

b 0 & 800 _____ b 200 & 800 _____ × 10^{-3} mm^2/s

3 ✎ **Predict** In which region is the difference between ADC (b 0 & 800) and ADC (b 200 & 800) biggest? Check on the Difference map.

☐ Brain tissue ☐ CSF ☐ Vessels + CSF

Then check (vs b 200 & 800): brain _____ % CSF _____ % vessels + CSF _____ %

4 A change in brain ADC is real only above **11 %** (QIBA). Scan 1 used b 0 & 800, the follow-up b 200 & 800, and the tissue did not change. What change would you measure, and how much of the 11 % limit does it use up? And if both scans used b 0 & 800?

Measured change: _____ % Share of the 11 %: ☐ almost none ☐ about half ☐ all or more

Both scans b 0 & 800, change: ☐ about 0 % ☐ about 5 % ☐ about 10 %

TAB 2 · TREATMENT (a simulated tumour: treatment cuts the blood fraction f from 0.25 to 0.08; the cells, D, do not change)

1 ✎ **Predict** After treatment, does the ADC fall, stay the same or rise? Circle your guess. Then press **Show the answers** and write the real change.

	Guess	Real change
b 0 & 800	☐ Falls ☐ Same ☐ Rises	%
b 200 & 800	☐ Falls ☐ Same ☐ Rises	%

2–3 Were your guesses right? ☐ Both ☐ One ☐ Neither The scanner's ADC (b 0 & 800) falls 19 %. With the 11 % limit, a clinician reads it as:

☐ Real change: response ☐ Real change: no response ☐ No real change

4 What really changed? (b 200 & 800) ☐ The cells (D) ☐ Only the blood (f) ☐ Both

Change with b 200 & 800: _____ %

5 **Bonus** Look at the b 50 & 800 row. Is a lowest b-value of 50 enough?

_____

TAB 3 · NOISE

1–2 ✎ **Predict** Which IVIM map shows clear anatomy, and which looks like noise?

Clear: ☐ D ☐ f ☐ D* Noise: ☐ D ☐ f ☐ D* Right? ☐ Yes ☐ No

3 **Bonus** CSF behind the midbrain: f = _____ Does that make sense? _____

4–5 **Bonus** Scan the same voxel 10 times. Which stays close to its true value?

SNR 50 (default): ☐ D ☐ f ☐ D* SNR 20: ☐ D ☐ f ☐ D*

Your checklist

before you trust an ADC · fill in at the debrief

1 · Which b-values? _____

2 · Whose ROI? _____

3 · Bigger than the repeatability? _____

4 · Geometrically trustworthy? _____

Brain data: E. Peterson, IVIM dataset, Figshare 2016, doi:10.6084/m9.figshare.3395704 (CC0). True-change limits: QIBA DWI Profile, Stage 3, 2022 (brain 11 %, prostate 47 %). Lowest b-value: Kooreman et al., Radiother Oncol 2020.